

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

Recency of Sources

Students should prioritize recent and relevant literature and studies published within the last six (6) years, generally covering 2021 to 2026 for studies conducted in 2026.

The six-year recency requirement applies particularly to:

- Recent empirical studies;
- Artificial Intelligence and Generative AI technologies;
- Machine Learning and Data Analytics applications;
- Business Intelligence technologies;
- Internet of Things and Smart Technologies;
- Current system implementations and applications; and
- Recent evaluation approaches and technological developments.

However, seminal, foundational, or standard references may be older than six years when necessary. These may include:

- Original or foundational algorithm publications;
- Established theories and frameworks;
- International standards such as ISO/IEC standards;
- Widely accepted evaluation instruments such as the System Usability Scale (SUS);
- Foundational technological concepts.

Students should avoid relying heavily on outdated sources when recent research is available. The review should demonstrate the current state of research and technological development.

Purpose of Chapter 2

Chapter 2 presents a **critical and integrated review of related literature and studies** relevant to the proposed research. Its purpose is to establish the conceptual, technical, and empirical foundation of the study by examining existing knowledge, technologies, methods, algorithms, systems, and evaluation approaches.

The chapter should demonstrate how previous researchers and authors have addressed problems related to the proposed study. It should identify established findings, commonly used approaches, differences among studies, limitations of existing implementations, and areas that require further investigation.



Related literature and related studies should be integrated within each thematic discussion. Students should not necessarily create separate sections entitled *Related Literature* and *Related Studies* if both types of sources address the same research theme. Instead, each theme should combine conceptual literature, technical literature, and empirical studies to provide a comprehensive discussion.

The review should follow a logical progression from the **research domain and problem context** toward the **data, algorithms or models, system/application, and evaluation approaches** relevant to the study.

Recommended Structure of Chapter 2

- A. Introduction to the Review
- B. Review of Related Literature and Studies

The themes should generally follow:

1. Domain and Problem Context
2. Data and Related Approaches
3. Algorithms, Models, and Technologies
4. Existing Systems and Applications
5. Evaluation Approaches

Within every theme, students should integrate:

Related Literature + Related Studies + Comparison/Synthesis

The exact subthemes may vary depending on the nature of the research.

- C. Synthesis of the Reviewed Literature and Studies
- D. Research Gap and Study Contribution
- E. Conceptual Framework
- F. Closing Paragraph

General Student Guide for Each Theme

Each thematic section should not simply define a technology or summarize individual studies. Students should write the discussion as a connected narrative.

A recommended writing pattern is:

Paragraph 1 – Establish the Concept or Context

Introduce the major concept or problem addressed by the theme using recent and credible literature.

Paragraph 2 – Discuss Existing Approaches

Explain the technologies, methods, models, or approaches discussed in related literature.

Paragraph 3 – Integrate Empirical Studies



Discuss how recent studies have applied, implemented, compared, or evaluated the approaches.

Paragraph 4 – Compare and Connect

Identify similarities, differences, strengths, limitations, or trends among the reviewed literature and studies, and explain their relevance to the proposed research.

Students do not need to follow four paragraphs for every theme rigidly. The number of paragraphs should depend on the complexity and amount of available literature.

However, the discussion should demonstrate the following progression:

Concept → Existing Knowledge → Empirical Evidence → Comparison → Relevance

EXAMPLE 1

Artificial Intelligence, Machine Learning, and Generative AI: AI-Powered Student Academic Risk Prediction and Advisory System

Introduction to the Review Student Guide

The introduction should identify the major areas that will be reviewed and explain their relevance to the proposed study. For this research, the review should cover student academic risk, educational data, Machine Learning-based prediction, Generative AI-supported advisory applications, existing AI-based systems, and evaluation approaches.

This chapter presents an integrated review of related literature and studies relevant to the development of an AI-powered student academic risk prediction and advisory system. The review examines student academic risk and early identification, the use and preparation of educational data for predictive analytics, Machine Learning approaches for academic risk classification, and the emerging use of Generative AI for personalized educational support. It also examines existing AI-based decision-support applications and the approaches used to evaluate predictive models and software systems. The reviewed literature and



studies provide the conceptual, technical, and empirical foundation for the proposed research and support the identification of the research gap addressed by the study.

Theme 1: Student Academic Risk and Early Identification

Student Guide

This theme should integrate literature discussing academic underperformance and early intervention with recent empirical studies that have attempted to identify students at risk. Students should discuss the indicators commonly associated with academic risk and compare how different studies define or measure risk.

Recent research demonstrates the continuing importance of early identification of students who may experience academic difficulties. Student performance prediction has become a major area of educational data mining and learning analytics because early predictions can support timely intervention and evidence-based academic decision-making. A recent systematic review by Shi et al. (2025) found that academic and behavioral engagement factors are among the most commonly used predictors in Machine Learning studies involving at-risk students, although definitions of academic risk vary across studies.

Similarly, a comprehensive review of Machine Learning and Deep Learning approaches reported that student performance prediction research commonly relies on educational datasets containing academic and engagement-related indicators and emphasizes the importance of connecting predictive results with meaningful educational interventions (Predicting Student Performance, 2026). These findings suggest that the identification of academic



risk should consider the specific characteristics and available data of the institution in which the predictive system will be implemented.

Theme 2: Educational Data and Data Preparation

Student Guide

This theme should support Specific Objective 1. The discussion should examine the types of data used in previous studies and issues involving data quality, feature selection, preprocessing, class imbalance, and data relevance.

The quality and relevance of educational data substantially influence the performance and reliability of academic prediction models. Recent reviews show that Machine Learning studies commonly use academic performance and student engagement indicators, although the selection of predictors differs by learning environment and research context (Shi et al., 2025). A recent state-of-the-art review of student success prediction likewise reported that prior academic performance and other student-related factors are frequently considered when developing predictive models, while the selection of variables and evaluation procedures varies considerably across studies (Predicting Student Success, 2026).

These findings demonstrate that data preparation should not be treated as a purely technical activity. Researchers must determine which variables are relevant, address incomplete or inconsistent records, and ensure that the resulting dataset appropriately represents the academic problem being investigated.

Theme 3: Machine Learning for Academic Risk Prediction



Student Guide

This theme should support Specific Objective 2. Students should integrate recent literature and empirical studies comparing different classification algorithms.

Possible algorithms may include:

- Logistic Regression;
- Decision Tree;
- Random Forest;
- Support Vector Machine;
- XGBoost;
- Neural Networks.

Machine Learning approaches have been extensively applied to student performance and academic risk prediction. Recent systematic reviews indicate that tree-based models, neural networks, and other Machine Learning approaches are frequently used to analyze complex educational datasets and identify students who may require academic intervention (Predicting Student Performance, 2026). Shi et al. (2025) further observed an increasing use of deep learning and ensemble approaches in at-risk student prediction, while also noting concerns related to reproducibility, generalizability, and model interpretability.

Recent research likewise identifies Random Forest, Support Vector Machine, Decision Tree, Logistic Regression, and Artificial Neural Networks as frequently applied approaches in student performance prediction (Predicting Student Academic Performance, 2026). These findings indicate that no single algorithm can automatically be considered the most appropriate for every institutional dataset. Comparative implementation using appropriate performance measures is therefore necessary to determine which model best supports the academic risk prediction task of the proposed study.



Theme 4: Generative AI and Personalized Academic Support

Student Guide

This theme should discuss recent literature and empirical studies involving:

- Personalized feedback;
- AI-generated recommendations;
- Adaptive learning;
- Educational assistance;
- Academic advisory support.

Students should also discuss concerns regarding accuracy, bias, privacy, and human oversight.

Generative AI has introduced new possibilities for providing personalized feedback and educational support. A systematic review of Machine Learning and Generative AI applications in higher education found that while Machine Learning remains dominant in predictive learning analytics tasks, Generative AI is increasingly being explored for adaptive feedback, personalized learning, and other forms of educational assistance (Machine Learning and Generative AI in Learning Analytics for Higher Education, 2025).

Recent empirical research has also demonstrated the potential of Generative AI to generate context-sensitive feedback using learner-related information (Qian et al., 2026). Similarly, recent research on Generative AI feedback in higher education has examined how AI-generated feedback may support student learning while emphasizing the importance of understanding learner interaction with the generated responses (Supporting Self-Regulated Learning, 2026). These findings suggest that Generative AI can extend the usefulness of predictive systems by transforming analytical results into



understandable and personalized recommendations, although generated outputs should remain subject to appropriate human oversight.

Theme 5: AI-Based Decision-Support and Advisory Systems

Student Guide

This theme should review systems that combine:

- Educational data;
- Predictive analytics;
- Risk classification;
- Dashboards;
- Recommendations;
- Decision-support mechanisms.

Recent developments in learning analytics demonstrate a movement toward integrating predictive models with educational support mechanisms rather than treating prediction as an isolated analytical task. The systematic review of Machine Learning and Generative AI applications in higher education reported that predictive models are widely used for engagement and dropout-risk analysis, while Generative AI applications are emerging in adaptive feedback and personalized support (Machine Learning and Generative AI in Learning Analytics for Higher Education, 2025).

Qian et al. (2026) further demonstrated an integrated framework combining learner-related data, intelligent analysis, and feedback generation for personalized educational assessment. However, recent literature also indicates that the practical value of AI-generated feedback depends on the educational



context, the quality of the generated content, and the manner in which learners and educators interact with the system (Generative AI-Driven Feedback in Higher Education, 2026). These findings support the integration of predictive academic risk classification with an advisory mechanism that assists authorized users in interpreting prediction results and considering possible intervention strategies.

Theme 6: Evaluation of AI Models and Systems

Student Guide

This theme should support Specific Objective 4.

The review should include evaluation approaches for:

Algorithms

- Accuracy;
- Precision;
- Recall;
- F1-score;
- Confusion Matrix.

System

- ISO/IEC 25010;
- System Usability Scale (SUS), when applicable.

Recent reviews of student performance prediction emphasize the use of multiple evaluation metrics because different measures provide different perspectives on predictive performance. Accuracy, precision, recall, and F1-score are among the commonly reported measures in studies involving student outcome prediction (Predicting Student Success, 2026). Recent systematic reviews also emphasize concerns regarding generalizability and reproducibility, demonstrating that strong performance on a particular dataset does not necessarily guarantee effectiveness in another educational context (Shi et al., 2025).



At the application level, predictive systems must also be assessed according to their practical usability and software quality. Therefore, the proposed study should evaluate both the performance of the selected Machine Learning models and the quality and usability of the developed academic advisory system.

EXAMPLE 2

Data Analytics and Business Intelligence: Data Analytics-Based Enrollment Forecasting and Decision Support System

This chapter presents an integrated review of related literature and studies relevant to the development of a data analytics-based enrollment forecasting and decision support system. The review examines enrollment management and institutional planning, the use of historical data for predictive analytics, forecasting models for time-dependent data, and Business Intelligence technologies for presenting analytical information. It also reviews existing forecasting and decision-support applications and the approaches used to evaluate predictive models and software systems. The reviewed literature and studies provide the conceptual and empirical basis for the proposed research and support the identification of its research gap and contribution.

Theme 1: Enrollment Management and Institutional Planning

Student Guide

This theme should integrate recent literature concerning higher education planning with empirical studies involving enrollment analysis and forecasting.

Enrollment information is an important component of higher education planning because changes in student population may influence academic program management, faculty requirements, facilities, and resource allocation.



Recent forecasting research has increasingly applied data-driven approaches to support institutional decision-making. For example, Sembiring et al. (2026) compared Simple Moving Average, ARIMA, and Random Forest approaches for forecasting student admissions across multiple academic programs, demonstrating the value of comparing different approaches rather than relying on a single forecasting technique. A recent Philippine study by Dormendo and Maliberan (2026) also applied Artificial Neural Network-based time-series forecasting to enrollment data from a higher education institution, highlighting the role of predictive models in supporting academic planning. These studies demonstrate that institutional enrollment data can provide a valuable basis for forecasting, although the suitability of a particular model depends on the characteristics of the local dataset.

Theme 2: Enrollment Data and Data Analytics

Student Guide

This theme should support **Specific Objective 1**. Review literature concerning historical data, temporal data, preprocessing, trend analysis, and data quality.

Data analytics provides approaches for transforming historical institutional records into information that supports planning and forecasting. Enrollment forecasting studies commonly rely on time-dependent historical observations and may incorporate program-level or institutional variables depending on the available dataset. The recent study of Sembiring et al. (2026) demonstrates the use of historical admissions data across multiple study programs to compare forecasting approaches, while Dormendo and Maliberan (2026) illustrate the



application of institutional enrollment records in developing a Machine Learning-based forecasting model.

These studies highlight the importance of preparing historical records and identifying the temporal structure of the data before model implementation. Missing observations, changes in academic programs, and inconsistencies in historical records may affect the reliability of forecasting results.

Theme 3: Forecasting Models and Predictive Analytics

Student Guide

This theme should support Specific Objective 2.

Students may discuss and compare:

- ARIMA;
- SARIMA;
- Random Forest Regression;
- XGBoost;
- LSTM;
- Artificial Neural Networks.

Recent studies demonstrate the continued use of both statistical and Machine Learning approaches for higher education enrollment forecasting. Sembiring et al. (2026) directly compared traditional methods and a Machine Learning approach, including SMA, ARIMA, and Random Forest, in forecasting student admissions. Dormendo and Maliberan (2026) similarly explored an Artificial Neural Network approach for predicting enrollment trends in a Philippine higher education context.

These studies demonstrate that different models provide alternative ways of representing temporal patterns and relationships within historical enrollment data. Their findings also support the need for comparative model evaluation



because the performance of forecasting approaches may differ according to data volume, temporal patterns, and institutional conditions.

Theme 4: Business Intelligence and Decision-Support Applications

Student Guide

This theme should review recent literature and studies involving:

- Institutional analytics;
- Dashboards;
- Data visualization;
- Forecast presentation;
- Decision support.

Business Intelligence extends the value of data analytics by presenting analytical results through dashboards, reports, and visualizations that can support interpretation and decision-making. In an institutional forecasting context, predictive results become more useful when decision-makers can examine historical trends and projected values in an accessible form. Recent enrollment forecasting studies such as those of Sembiring et al. (2026) and Dormendo and Maliberan (2026) demonstrate the potential value of predictive information for higher education planning, although forecasting research often focuses primarily on model development and predictive accuracy.

This creates an opportunity for system-development research to integrate forecasting capabilities with interactive visualization and decision-support features. The proposed study therefore extends the analytical component by presenting enrollment trends and forecasts through a system designed for authorized institutional users.



Theme 5: Evaluation of Forecasting and Decision-Support Systems

Student Guide

This theme should support Specific Objective 4.

Discuss:

Forecasting Models

- MAE;
- RMSE;
- MAPE.

System

- ISO/IEC 25010;
- SUS, when applicable.

Forecasting research commonly evaluates models by measuring the difference between predicted and actual values. Recent Machine Learning-based forecasting studies in institutional contexts emphasize the need to compare model performance using quantitative error measures before selecting an approach for practical use (Sembiring et al., 2026; Dormendo & Maliberan, 2026).

However, predictive performance alone does not determine the usefulness of a decision-support application. Users must also be able to access, understand, and apply the information generated by the system. Therefore, the proposed study should evaluate both the forecasting performance of the selected model and the relevant quality and usability characteristics of the developed decision-support application.

EXAMPLE 3

Internet of Things, Embedded Systems, and Smart Technologies: Smart Campus Energy Monitoring and Predictive Control System Using IoT Sensors

This chapter presents an integrated review of related literature and studies relevant to the development of a smart campus energy monitoring and predictive



control system using Internet of Things technologies. The review examines smart campus energy management, IoT and embedded systems for sensor-based data collection, predictive analytics for energy consumption forecasting, and existing smart monitoring applications. It also examines the methods used to evaluate predictive models and the quality and usability of IoT-based systems. The reviewed literature and studies provide the conceptual, technical, and empirical foundation for the proposed research and guide the identification of the research gap addressed by the study.

Theme 1: Smart Campus and Energy Management

Student Guide

This theme should integrate recent literature on smart environments and energy management with empirical studies involving campus or building energy monitoring.

Smart campus initiatives increasingly use digital and sensor-based technologies to improve the management of institutional facilities and resources. Energy management is particularly relevant because campus buildings and facilities may have varying patterns of electricity consumption that are difficult to understand through periodic manual readings alone. Recent research on campus building energy forecasting demonstrates the potential of Machine Learning to analyze historical and environmental information for sustainable energy management (Machine Learning-Based Forecasting of Campus Building Energy Consumption, 2026).

A recent study by Maderazo et al. (2026) also developed an IoT-enabled energy monitoring system that combined real-time energy data collection with



Machine Learning-based consumption analysis. These studies demonstrate the potential of integrating continuous monitoring and predictive analytics to support data-driven energy management.

Theme 2: IoT Data Acquisition and Embedded Systems

Student Guide

This theme supports Specific Objective 1.

Review:

- IoT sensors;
- Embedded devices;
- Data acquisition;
- Connectivity;
- Data transmission;
- Data quality and preprocessing.

The Internet of Things enables physical devices and sensors to collect and transmit information from real-world environments for monitoring and analysis. Recent IoT applications demonstrate how sensors and embedded devices can be integrated with centralized platforms to support continuous or near-real-time data collection. Maderazo et al. (2026) developed an IoT-enabled energy monitoring system that collected electrical usage data through connected energy meters and transmitted the information for analysis and prediction.

Similarly, a recent smart campus monitoring study integrated sensors, microcontrollers, cloud services, and predictive models to collect and forecast environmental data within academic environments (Smart Noise Monitoring and Forecasting, 2026). These studies demonstrate that the quality and continuity of sensor data are essential considerations because communication interruptions,



inaccurate measurements, and missing observations may affect subsequent predictive analysis.

Theme 3: Predictive Analytics for Energy Consumption

Student Guide

This theme supports Specific Objective 2.

Possible models include:

- Random Forest Regression;
- XGBoost;
- LSTM.

Students should discuss comparative studies and model performance.

Predictive analytics has become an important component of smart energy management because forecasting models can estimate future consumption based on historical and continuously collected data. Recent research on campus building energy forecasting compared Machine Learning approaches and reported that model performance varied across algorithms, with an LSTM model achieving strong short-term forecasting performance in the examined dataset (Machine Learning-Based Forecasting of Campus Building Energy Consumption, 2026).

Maderazo et al. (2026) likewise integrated Machine Learning techniques into an IoT-enabled energy monitoring system to identify consumption patterns and forecast future usage. These findings demonstrate that predictive models can extend the functionality of monitoring systems beyond simple visualization. However, model performance remains dependent on the characteristics and quality of the available data, supporting the comparative implementation and evaluation of selected forecasting algorithms in the proposed study.



Theme 4: IoT-Based Smart Monitoring Systems

Student Guide

This theme should review recent systems that integrate:

- Sensors;
- Embedded devices;
- Connectivity;
- Data storage;
- Dashboards;
- Real-time monitoring;
- Predictive analytics.

Recent IoT-based monitoring systems demonstrate the integration of sensing technologies, embedded devices, communication platforms, data processing, and predictive analytics within a unified application. Maderazo et al. (2026) developed a system that combined IoT-enabled energy monitoring with Machine Learning-based analysis to provide real-time consumption tracking and predictive information.

Current smart campus studies similarly integrated environmental sensors, ESP8266 microcontrollers, cloud-based data services, and a deep learning forecasting model to support monitoring and decision-making in an academic environment (Smart Noise Monitoring and Forecasting, 2026). These implementations demonstrate that smart monitoring systems can provide greater value when sensor data are not only collected but also processed, analyzed, and presented in a form that supports practical decision-making.

Theme 5: Evaluation of IoT and Predictive Systems

Student Guide

This theme supports Specific Objective 4.

Review evaluation approaches for:

Predictive Models

- MAE;



- RMSE;
- MAPE.

IoT System

- Functional suitability;
- Performance efficiency;
- Reliability;
- Usability;
- ISO/IEC 25010;
- SUS, when applicable.

Recent IoT and predictive analytics studies demonstrate the importance of evaluating both analytical performance and the functionality of the integrated system. The recent campus building energy forecasting study evaluated model performance using quantitative forecasting measures and reported differences in performance among Machine Learning approaches (Machine Learning-Based Forecasting of Campus Building Energy Consumption, 2026).

Similarly, smart campus monitoring research has used predictive performance measures such as MAPE to assess forecasting capability while demonstrating the integration of sensors, communication technologies, and monitoring platforms (Smart Noise Monitoring and Forecasting, 2026). These findings indicate that the proposed study should evaluate both the accuracy of the selected predictive models and the quality of the developed IoT-based application.

Literature Support per Theme

Each major theme should be supported by multiple related literature and studies. Students should avoid relying on a single source to support an entire thematic discussion.

As a general guide:



Component	Recommended Practice
Major Theme	At least 3–5 relevant sources
Related Literature	Integrated with empirical studies
Recent Sources	Primarily 2021–2026
Seminal Sources	Allowed when necessary
Synthesis	Compare multiple sources
Research Gap	Must emerge from reviewed literature

NOTE: The sample statements per theme (2 paragraphs) are just compressed for presentation but should comply with the minimum RRL requirements. 30 RRLs

Important note to students:

1. Retrieve and read the full article, not only the abstract.
2. Verify the complete author names and bibliographic details.
3. Use the university's required APA edition consistently.
4. Ensure that every in-text citation appears in the Reference List.
5. Avoid citing sources you have not personally reviewed.
6. Prioritize peer-reviewed and indexed scholarly sources when available.

Synthesis

The Synthesis section integrates and critically examines the major findings from the related literature and studies reviewed in the preceding sections. Unlike a simple summary, the synthesis should connect findings across different themes, identify common patterns and differences, discuss the strengths and limitations of existing approaches, and explain how the reviewed evidence informs the proposed study.

The synthesis should help answer the following questions:

- What are the major findings across the reviewed literature and studies?
- What common approaches, technologies, or methods emerge?
- Where do the findings agree or differ?
- What limitations or unresolved issues remain?
- How do these findings support the direction of the proposed study?

The synthesis should serve as a transition from the Review of Related Literature and Studies to the Research Gap and Study Contribution. While the synthesis may identify limitations and unresolved issues, the next section should clearly define the specific research gap addressed by the study.



Guide for Writing the Synthesis

Identify the Major Patterns Across the Review

The student should examine the reviewed themes collectively rather than discussing each source separately. Identify recurring findings related to the research domain, data, algorithms or models, system development, and evaluation. For example, the literature may consistently demonstrate the value of predictive analytics while also showing differences in the algorithms used and the datasets on which they are evaluated.

Integrate Literature and Studies

The synthesis should combine conceptual literature and empirical studies. Students should avoid repeating individual summaries such as “*Author A found...*” followed by “*Author B found...*”. Instead, they should explain the overall pattern emerging from multiple sources.

For example:

Recent literature and empirical studies consistently demonstrate the potential of Machine Learning to improve predictive decision-making. However, the performance of individual algorithms varies according to dataset characteristics, feature selection, and the context in which the models are applied.

Identify Agreements and Differences

Students should discuss areas where the reviewed literature and studies reach similar conclusions and areas where findings differ. Differences may arise because of variations in datasets, research settings, algorithms, technologies, sample sizes, or evaluation metrics.

Students should not assume that one technology or algorithm is universally superior simply because it performed well in one study.

Connect the Findings to the Proposed Study

The synthesis should explain how the collective findings guide the proposed study. The discussion should show how previous research supports the selection of data, algorithms, technologies, system features, and evaluation approaches.

The synthesis should maintain alignment with the study objectives:

Data → Algorithms/Models → System/Application → Evaluation

Identify Remaining Issues

The student should identify recurring limitations or unresolved concerns that emerge from the literature and studies. These may include:



- Limited datasets;
- Context-specific findings;
- Lack of algorithm comparison;
- Limited system integration;
- Absence of personalized features;
- Insufficient system evaluation;
- Privacy, security, or ethical concerns;
- Limited implementation in a particular institutional or local context.

These issues will provide the foundation for the succeeding **Research Gap and Study Contribution** section.

Write as an Integrated Narrative

The synthesis should generally be written in a series of connected paragraphs. Subtitles may be used as writing guides during drafting, but students should follow the required format of the thesis manual when preparing the final manuscript.

A recommended flow is:

Overall Findings → Data and Context → Algorithms/Technologies → System/Application → Evaluation and Challenges → Connection to the Proposed Study

EXAMPLE 1

Artificial Intelligence, Machine Learning, and Generative AI: AI-Powered Student Academic Risk Prediction and Advisory System

Student Guide

The synthesis should integrate the findings related to student academic risk, educational data, Machine Learning prediction models, Generative AI-supported advisory systems, and evaluation approaches. The discussion should demonstrate that predictive analytics can support early intervention but that model performance and practical usefulness depend on institutional data, algorithm selection, system integration, and appropriate evaluation.

The reviewed literature and studies demonstrate the increasing role of Artificial Intelligence and Machine Learning in supporting data-driven



decision-making in education. A consistent finding across the reviewed works is that educational data can be analyzed to identify patterns associated with student academic performance and potential academic risk. Academic indicators, assessment results, attendance, and other relevant student-related information have been used to develop predictive models that support the early identification of students who may require intervention. However, the reviewed studies also demonstrate that the selection and availability of relevant data vary across educational institutions. This suggests that the identification and preparation of institution-specific academic data remain essential before implementing predictive models.

Another major theme emerging from the review is the increasing use of Machine Learning algorithms for academic risk prediction. Different studies have applied and compared algorithms such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, gradient-boosting models, and neural networks. Although several studies report strong performance from particular algorithms, the findings do not establish a single model as universally superior. Model performance varies according to the characteristics of the dataset, the selected predictors, class distribution, preprocessing procedures, and institutional context. The reviewed literature therefore supports the comparative implementation and evaluation of selected Machine Learning models rather than assuming that an algorithm that performs well in one educational environment will automatically produce similar results in another.



The review also demonstrates the emerging potential of Generative AI to complement predictive analytics. While Machine Learning models primarily identify patterns and generate predictions, Generative AI can provide natural-language explanations, personalized feedback, and context-sensitive recommendations. Recent developments suggest that these technologies can extend the practical usefulness of analytical systems by helping users interpret predictive results and identify possible actions. However, the literature also highlights concerns involving inaccurate generated content, bias, privacy, and the need for appropriate human oversight. These findings indicate that Generative AI should support rather than independently replace academic decision-making.

Across the reviewed literature and studies, another important finding is that predictive performance alone is insufficient to determine the effectiveness of an AI-based educational application. Prediction models must be evaluated using appropriate performance measures, including accuracy, precision, recall, and F1-score. In academic risk identification, recall may be particularly important because failing to identify students who require intervention may reduce the value of the predictive system. At the same time, the developed application must be evaluated in terms of its functionality, usability, and other relevant software quality characteristics to determine whether intended users can effectively access and utilize the generated information.

Overall, the reviewed literature establishes that AI technologies can support the early identification of academic risk and provide opportunities for more personalized educational support. The literature collectively supports the



use of relevant academic data, the comparison of appropriate Machine Learning algorithms, the development of an integrated prediction and advisory application, and the evaluation of both predictive models and the developed system. However, differences in institutional datasets, algorithm performance, and the integration of Generative AI into practical advisory applications indicate the need for context-specific research. These insights provide the foundation for the proposed study, which seeks to integrate academic risk prediction with Generative AI-supported intervention recommendations within a system designed to support data-driven academic decision-making.

EXAMPLE 2

Data Analytics and Business Intelligence: Data Analytics-Based Enrollment Forecasting and Decision Support System

Student Guide

The synthesis should integrate findings concerning enrollment management, historical institutional data, forecasting models, Business Intelligence tools, and system evaluation. The discussion should demonstrate the importance of using data for institutional planning while recognizing that forecasting performance depends on the characteristics of the available data and the selected model.

The reviewed literature and studies demonstrate the importance of data analytics in supporting evidence-based planning and decision-making in higher education institutions. A common finding is that historical enrollment data can provide valuable information about trends and patterns that may assist administrators in anticipating future student populations. Enrollment forecasting can support decisions related to academic program planning, faculty requirements, facilities, and resource allocation. However, the reviewed studies also demonstrate that institutional enrollment patterns may be influenced by



changes in academic programs, policies, demographic conditions, and other contextual factors. This indicates that forecasting systems should be developed using data that accurately represent the institutional environment in which they will be implemented.

Another major finding concerns the importance of data preparation and analysis before forecasting models are implemented. Historical enrollment records must be examined for completeness, consistency, and temporal patterns because missing observations and changes in data structures may affect forecasting results. The literature further demonstrates that descriptive and predictive analytics serve complementary purposes. Descriptive analytics helps decision-makers understand historical enrollment trends, while predictive analytics provides estimates of possible future outcomes. The integration of these approaches can therefore provide a more comprehensive basis for institutional decision-making.

The reviewed studies also demonstrate the use of different forecasting approaches, including statistical time-series models and Machine Learning or deep learning techniques. Models such as ARIMA and SARIMA are commonly used to analyze temporal and seasonal patterns, while approaches such as Random Forest, XGBoost, and Artificial Neural Networks can model more complex relationships within the data. The findings across the studies differ regarding which model produces the best forecasting performance. These differences indicate that forecasting accuracy is influenced by data characteristics, the length of historical records, seasonality, and the forecasting



period. Consequently, the literature supports the comparison of multiple forecasting models using appropriate performance measures rather than relying on a single method.

Business Intelligence also emerges as an important component in transforming analytical results into information that can be understood and used by decision-makers. The reviewed literature demonstrates that forecasting outputs become more valuable when they are presented through dashboards, visualizations, and accessible reports. While many forecasting studies concentrate primarily on algorithm development and predictive accuracy, fewer system-oriented approaches emphasize the integration of forecasting models with interactive decision-support functionalities. This demonstrates the importance of connecting the analytical component with an application that enables authorized users to examine historical trends and projected enrollment information.

The review further emphasizes that the evaluation of a forecasting-based decision-support system should address both the analytical model and the developed application. Forecasting models should be assessed using appropriate error measures such as Mean Absolute Error, Root Mean Squared Error, and Mean Absolute Percentage Error. At the system level, relevant software quality characteristics and usability should also be evaluated to determine whether the application effectively supports its intended users. A forecasting model with acceptable predictive performance may still have limited



practical value if the resulting information cannot be easily accessed, understood, or used by institutional decision-makers.

Overall, the reviewed literature and studies establish that data analytics and forecasting can provide valuable support for enrollment management and institutional planning. The collective findings support the systematic preparation of historical enrollment data, the comparative evaluation of forecasting models, and the integration of analytical results within a Business Intelligence-based decision-support application. However, differences in forecasting performance across datasets and the limited integration of predictive models with institution-specific decision-support features demonstrate the need for context-based implementation and evaluation. These findings guide the proposed study in developing a system that combines enrollment forecasting, data visualization, and decision-support functionalities within a unified application.

EXAMPLE 3

Internet of Things, Embedded Systems, and Smart Technologies: Smart Campus Energy Monitoring and Predictive Control System Using IoT Sensors

Student Guide

The synthesis should integrate the findings concerning smart campus environments, IoT-based data acquisition, embedded systems, predictive analytics, monitoring applications, and system evaluation. It should demonstrate how sensor technologies and predictive models can work together while recognizing technical challenges involving data reliability, connectivity, and system performance.

The reviewed literature and studies demonstrate the growing role of Internet of Things technologies in the development of smart environments and



resource management systems. A common finding is that IoT sensors and embedded devices enable the continuous collection of information from physical environments, providing data that may not be available through traditional manual monitoring methods. In energy management applications, sensor-based systems can capture measurements related to electricity consumption and transmit these data to centralized platforms for analysis. These developments demonstrate the potential of IoT technologies to support more timely and data-driven monitoring of institutional facilities.

The review also highlights the importance of reliable data acquisition and preprocessing in IoT-based applications. Although sensors and embedded devices enable continuous or near-real-time monitoring, the quality of the collected information may be affected by inaccurate readings, missing observations, communication interruptions, and network limitations. The reviewed studies demonstrate that predictive analysis depends heavily on the quality and consistency of sensor-generated data. Consequently, the systematic identification, collection, validation, and preparation of IoT data are necessary before the information can be effectively used for analytical and predictive purposes.

Another major theme emerging from the literature is the increasing integration of Machine Learning and predictive analytics into smart monitoring systems. Traditional monitoring applications primarily collect and display current information, whereas predictive approaches extend system functionality by estimating future conditions or consumption patterns. The reviewed studies have



applied different models, including tree-based Machine Learning approaches, gradient-boosting algorithms, and Long Short-Term Memory networks, to energy forecasting tasks. However, the findings demonstrate that model performance varies according to data frequency, historical data volume, selected features, and environmental conditions. This indicates that the selection of a predictive model should be based on comparative evaluation using the actual data collected within the intended implementation environment.

The reviewed studies further demonstrate that the value of IoT-based applications increases when multiple technologies are integrated within a unified system. Smart monitoring systems may combine sensors, embedded devices, communication technologies, data storage platforms, predictive models, and dashboards to provide more comprehensive information to users. However, existing implementations differ in the extent to which these components are integrated. Some systems emphasize real-time monitoring, while others focus primarily on predictive analysis. This variation supports the development of applications that combine data acquisition, monitoring, analysis, and predictive capabilities within a single system.

The literature also emphasizes the importance of evaluating both the predictive models and the developed IoT application. Forecasting performance should be assessed using appropriate error measures, such as Mean Absolute Error, Root Mean Squared Error, and Mean Absolute Percentage Error. At the system level, relevant quality characteristics such as functional suitability, performance efficiency, reliability, and usability should be examined. This dual



evaluation is necessary because an accurate predictive model alone does not guarantee the effectiveness of the overall system. The monitoring application must also reliably collect, process, and present information to its intended users.

Overall, the reviewed literature and studies establish that IoT technologies and predictive analytics can contribute to more intelligent and data-driven energy management. The collective findings support the use of sensor-generated data, the comparative implementation of predictive models, and the integration of monitoring and analytical functions within a smart application. However, recurring challenges involving sensor reliability, data quality, network connectivity, predictive performance, and system integration remain important considerations. These findings provide the foundation for the proposed study, which seeks to integrate IoT-based energy monitoring and predictive analytics within a smart campus application designed to support more informed energy management and decision-making.

General Synthesis Format for Students

Students may use the following sentence-flow guide when writing their own synthesis:

- Paragraph 1 – Overall Findings:
Begin by presenting the overall trend that emerges from the reviewed literature and studies. Explain what the collective body of research generally demonstrates about the research problem or domain
- Paragraph 2 – Data and Context:
Synthesize findings concerning the data, variables, information sources, or contextual factors required by the study. Discuss common findings and important differences across studies.
- Paragraph 3 – Algorithms, Models, or Technologies:
Integrate findings regarding the algorithms, models, or technologies used



in previous studies. Explain similarities and differences in performance and why comparative evaluation may be necessary.

□ Paragraph 4 – System or Application:

Discuss how existing studies have translated technologies or analytical models into systems, applications, platforms, or decision-support tools. Identify common functionalities and limitations.

□ Paragraph 5 – Evaluation and Challenges:

Synthesize the approaches used to evaluate algorithms and developed systems. Discuss common challenges, limitations, or concerns identified across the literature.

□ Final Paragraph – Connection to the Proposed Study:

Summarize how the collective findings inform the proposed research. Explain how the study will be guided by the reviewed evidence without yet presenting the complete research gap in detail.

Important Reminder for Students

The Synthesis is not a repetition of the Review of Related Literature and Studies. Avoid writing:

Study A used Random Forest. Study B used SVM. Study C used XGBoost.
Instead, synthesize:

The reviewed studies demonstrate that different Machine Learning algorithms have been applied to the prediction task, but their performance varies according to the characteristics of the dataset and the selected predictors. This supports the comparative evaluation of multiple algorithms in the proposed study.

The desired outcome is a connected interpretation of the entire review, following the established Chapter 2 progression:

Related Literature and Studies → Synthesis → Research Gap → Study Contribution

This structure also maintains alignment with the study objectives:

Data → Algorithms/Models → System/Application → Evaluation

Gap Bridge of the Study

The Gap Bridge of the Study explains the connection between what is already known from the reviewed literature and studies and what still needs to be



investigated. It identifies the specific limitations, inconsistencies, underexplored contexts, or missing integrations in previous research and explains how the proposed study is positioned to address them.

The Gap Bridge should not simply claim that “there are no studies about the topic.” Instead, students should demonstrate the gap by carefully examining what previous studies have already accomplished and identifying what remains insufficiently addressed.

The Gap Bridge serves as the transition between the Synthesis of Literature and Studies and the proposed study. It should clearly answer:

- What has previous research already established?
- What remains limited, inconsistent, or underexplored?
- Why is the remaining issue important?
- How will the proposed study address it?
- What specific contribution will the study make?

The discussion should remain aligned with the study's objectives:

Data → Algorithms/Models → System/Application → Evaluation

Guide for Writing the Gap Bridge

The student should begin by connecting the discussion to the major findings presented in the synthesis. Briefly establish what previous literature and studies have demonstrated about the research domain. The purpose is not to repeat the entire review but to provide enough context to show that the proposed gap is grounded in existing research.

The student should then identify the specific gap that emerges from the reviewed literature. A research gap may involve an underexplored context, limited or unavailable local data, inconsistent findings regarding algorithms or models, the lack of integration between technologies, insufficient evaluation, or a practical problem that existing systems have not adequately addressed. The gap must be based on the reviewed literature and studies rather than on an unsupported assumption.

After identifying the gap, the student should explain why addressing it is necessary. The discussion should establish the relevance of the gap to the intended users, organization, community, industry, or research domain.

The student should then explain how the proposed study will address the identified gap. This should be written carefully. Students should avoid claiming that the study will completely solve a broad problem. Instead, they should explain the specific contribution of the study based on its objectives.



The Gap Bridge should normally end by stating the contribution of the study. The contribution may involve the development of a context-specific dataset or analytical approach, the comparison of selected algorithms, the integration of technologies into a functional system or application, or the comprehensive evaluation of both the algorithms and the developed system.

Common Types of Research Gaps in Computing Studies

Students may identify one or more of the following gaps, provided that these are supported by the reviewed literature and studies:

- Contextual Gap. Existing studies may have been conducted in different countries, industries, organizations, or environments, while the intended context remains underexplored.
- Data Gap. Previous studies may use datasets that differ substantially from the data available in the intended setting, or there may be limited research using local or institution-specific data.
- Algorithm or Model Gap. Previous studies may use different algorithms, but their comparative performance within the intended context remains unclear.
- Technology Integration Gap. Individual technologies may have been studied separately, while their integration into a unified application remains limited.
- System Development Gap. Previous studies may focus mainly on models or prototypes without developing a complete application for intended users.
- Evaluation Gap. Previous research may focus on algorithm performance but provide limited evaluation of software quality, usability, or real-world applicability.

Students should avoid forcing all possible gaps into the study. Only gaps that are clearly supported by the literature and are relevant to the study objectives should be included.

EXAMPLE 1

Artificial Intelligence, Machine Learning, and Generative AI: AI-Powered Student Academic Risk Prediction and Advisory System

Student Guide

For this study, the Gap Bridge should connect the reviewed literature on student academic risk prediction, educational data analytics, Machine Learning algorithms, and Generative AI. The discussion should establish that previous



studies have demonstrated the effectiveness of predictive analytics but that gaps may remain in context-specific data, comparative model evaluation, the integration of prediction and advisory functions, and the joint evaluation of AI models and the developed application.

The reviewed literature and studies demonstrate that Artificial Intelligence and Machine Learning can support the early identification of students who may experience academic difficulties. Previous research has applied different academic indicators and predictive models, including Logistic Regression, Decision Trees, Random Forest, Support Vector Machines, gradient-boosting models, and neural networks, to classify or predict student academic outcomes. These studies generally demonstrate the potential of predictive analytics to support early intervention; however, the performance of the models varies across datasets, educational settings, selected predictors, and evaluation procedures.

Despite these developments, the reviewed studies indicate several areas that require further investigation. First, many predictive models are developed using datasets that are specific to a particular educational institution or research environment. Consequently, the applicability of previously reported models to another institutional context cannot be assumed. The availability, quality, and characteristics of student academic data may influence the predictive performance of Machine Learning algorithms. This creates a need to examine and prepare context-specific academic data and evaluate appropriate predictive models within the intended educational setting.

A second gap concerns the selection and comparison of Machine Learning algorithms. While previous studies have reported strong results using



particular models, the findings do not consistently identify one algorithm as the most effective for all academic risk prediction tasks. Some studies emphasize traditional Machine Learning approaches, while others demonstrate the potential of more advanced models. However, direct comparisons using the same institutional dataset and consistent evaluation measures remain necessary to determine which selected approach is most appropriate for a specific implementation context.

The review also demonstrates that many existing studies concentrate primarily on predictive performance. Although identifying students who may be academically at risk is valuable, prediction alone does not necessarily provide users with understandable or actionable support. The integration of Generative AI creates an opportunity to transform predictive outputs into personalized, human-readable advisory recommendations. However, the integration of Machine Learning-based risk prediction and Generative AI-supported academic advising within a unified system remains an area requiring further practical investigation, particularly with appropriate safeguards for responsible use and human oversight.

In addition, previous research often evaluates predictive models using performance metrics such as accuracy, precision, recall, and F1-score, while the usability and software quality of the developed application may receive less emphasis. A system intended to support academic decision-making should not only generate reliable predictions but should also provide information through a functional, understandable, and usable application.



In response to these gaps, the proposed study will examine relevant student academic data, implement and evaluate selected Machine Learning algorithms for academic risk prediction, and integrate the selected predictive approach with Generative AI-supported intervention recommendations in a unified application. The study will evaluate the predictive models using appropriate classification metrics, including precision, recall, and F1-score, while the developed system will be assessed using relevant software quality and usability measures.

Thus, the contribution of the study lies not only in developing another predictive model but in providing a context-specific, integrated, and evaluated AI-powered academic risk prediction and advisory system. The study seeks to contribute empirical evidence on the performance of selected Machine Learning approaches within the intended context while demonstrating how predictive analytics and Generative AI can be combined to support more informed and personalized academic intervention.

EXAMPLE 2

Data Analytics and Business Intelligence: Data Analytics-Based Enrollment Forecasting and Decision Support System

Student Guide

For this study, the Gap Bridge should connect the literature on enrollment analytics, historical institutional data, forecasting algorithms, Business Intelligence, and decision-support systems. The discussion should establish that enrollment forecasting has been widely studied but that forecasting performance remains context-dependent and that the integration of forecasting models into accessible decision-support applications may still require further investigation.



The reviewed literature and studies demonstrate that data analytics and forecasting techniques can support enrollment planning and institutional decision-making. Previous research has applied statistical time-series methods and Machine Learning approaches to analyze historical enrollment trends and estimate future student populations. These studies demonstrate that forecasting can provide useful information for planning academic programs, allocating resources, and anticipating institutional requirements.

However, the reviewed findings also show that enrollment forecasting performance varies across datasets and institutional contexts. Enrollment patterns may be influenced by academic policies, program offerings, demographic changes, and other local conditions. As a result, models that perform effectively using one institutional dataset may not necessarily produce the same level of accuracy when applied to another setting. This creates a need to examine and prepare historical enrollment data within the intended institutional context before selecting and implementing forecasting models.

Another gap concerns the comparative evaluation of forecasting approaches. Previous studies have applied models such as ARIMA, SARIMA, Random Forest, gradient-boosting algorithms, and neural networks. Although some approaches have demonstrated strong forecasting performance, the literature does not identify a universally superior method. Differences in data characteristics, forecasting periods, and evaluation procedures influence the reported results. Therefore, there remains a need to compare selected



forecasting models using the same institutional dataset and consistent error measures.

The reviewed studies further indicate that some research concentrates primarily on developing accurate forecasting models. While predictive accuracy is important, institutional decision-makers also require accessible tools for interpreting and using analytical results. Forecasts presented only as numerical outputs may have limited practical value for administrators who need to examine historical patterns, trends, comparisons, and projected enrollment information. The integration of forecasting models with Business Intelligence dashboards and decision-support features therefore remains important for translating analytical findings into usable institutional information.

An additional concern involves the evaluation of the complete application. Forecasting studies commonly assess models using error measures such as Mean Absolute Error, Root Mean Squared Error, and Mean Absolute Percentage Error. However, the quality and usability of the decision-support application may not always receive the same level of evaluation. A forecasting model may produce acceptable results but still provide limited institutional value if the information is difficult to access or interpret.

To address these gaps, the proposed study will identify and prepare relevant historical enrollment data, implement and compare selected forecasting algorithms, and develop a Data Analytics and Business Intelligence-based decision-support system for presenting enrollment trends and forecasts. The forecasting models will be evaluated using appropriate error measures, while the



developed system will be assessed based on relevant software quality and usability criteria.

The contribution of the study lies in integrating context-specific enrollment forecasting with Business Intelligence and decision-support functionalities within a unified application. By evaluating both the forecasting approaches and the developed system, the study seeks to provide practical evidence on how data analytics can support institutional planning and more informed decision-making within the intended organizational context.

EXAMPLE 3

Internet of Things, Embedded Systems, and Smart Technologies: Smart Campus Energy Monitoring and Predictive Control System Using IoT Sensors

Student Guide

For this study, the Gap Bridge should connect the literature on IoT monitoring, embedded systems, sensor-generated data, predictive analytics, smart energy management, and system evaluation. The discussion should demonstrate that existing research supports IoT-based monitoring but that challenges remain in integrating reliable data acquisition, predictive analytics, and practical decision-support features.

The reviewed literature and studies demonstrate the potential of Internet of Things technologies in supporting the development of smart monitoring and resource management systems. Previous research has shown that sensors and embedded devices can collect environmental and operational data continuously, allowing users to monitor conditions more efficiently than through traditional manual procedures. In energy management applications, IoT technologies have been used to collect electricity consumption data and provide information that supports monitoring and operational decision-making.



Despite these developments, the reviewed studies demonstrate that IoT-based systems remain highly dependent on the quality and reliability of sensor-generated data. Inaccurate readings, missing observations, communication interruptions, and network limitations may affect the information used for analysis. Since predictive models depend on historical and real-time data, the effectiveness of smart energy applications may be influenced by how sensor data are collected, validated, processed, and stored. This indicates the need for context-specific investigation of data acquisition and preparation within the intended implementation environment.

Another gap concerns the selection and evaluation of predictive models for smart energy applications. Previous studies have applied different approaches, including regression models, Random Forest, gradient-boosting algorithms, and Long Short-Term Memory networks. However, the findings vary depending on data frequency, environmental conditions, selected variables, and the amount of historical data available. Consequently, the literature does not establish a single predictive approach that is appropriate for every smart energy environment. Comparative evaluation using data obtained from the intended setting remains necessary.

The review further demonstrates that existing systems differ in the degree to which monitoring and predictive technologies are integrated. Some applications focus primarily on real-time data visualization, while others concentrate on forecasting future energy consumption. This separation may limit the practical usefulness of smart technologies when users need both current



information and predictive insights within a single application. There remains a need to investigate how IoT-based monitoring and predictive analytics can be integrated into a unified system that supports more comprehensive energy management.

The reviewed literature also indicates that system evaluation should extend beyond the predictive performance of the algorithms. Forecasting models may be assessed using measures such as Mean Absolute Error, Root Mean Squared Error, and Mean Absolute Percentage Error, but the overall IoT application must also be evaluated in terms of functional suitability, reliability, performance efficiency, and usability. Technical issues involving sensor reliability and network connectivity must likewise be considered when assessing the practical performance of the developed system.

To address these gaps, the proposed study will identify and collect relevant sensor-generated energy data, implement and compare selected predictive algorithms, and develop an IoT-based smart energy monitoring and predictive control application. The predictive models will be evaluated using appropriate forecasting error measures, while the developed application will be assessed using relevant software quality and usability criteria.

The contribution of the study lies in providing an integrated, context-specific, and evaluated IoT-based smart energy monitoring and predictive system. Rather than focusing solely on sensor monitoring or algorithmic forecasting, the study combines data acquisition, predictive analytics, monitoring, and decision-support functionalities within a unified application. The



findings may provide practical insights into the development and evaluation of smart technology solutions within similar institutional environments.

Recommended Writing Pattern for Students

Students may follow this paragraph sequence when writing the Gap Bridge:

- Paragraph 1 – What is Already Known.
Briefly state the major findings established by the reviewed literature and studies.
- Paragraph 2 – Data or Context Gap.
Explain whether existing data, datasets, populations, locations, or institutional contexts differ from the intended study.
- Paragraph 3 – Algorithm or Technology Gap.
Discuss inconsistencies, limited comparisons, or unresolved questions concerning the algorithms, models, or technologies.
- Paragraph 4 – System Integration Gap.
Identify whether previous studies focused on individual technologies without integrating them into a complete system or application.
- Paragraph 5 – Evaluation Gap.
Explain whether previous research focused primarily on algorithm performance while providing limited evaluation of the developed application.
- Paragraph 6 – How the Study Bridges the Gap.
State how the proposed study will address the identified gaps according to the objectives.
- Final Paragraph – Study Contribution.
Clearly explain the specific contribution of the study without overstating its expected impact.

Important Reminder

A strong Gap Bridge should be based on evidence from the reviewed literature and studies. Students should avoid unsupported statements such as:

- “No study has ever developed a system like this.”

Instead, use evidence-based language such as:



- “Although previous studies have demonstrated the effectiveness of predictive models, limited research has examined their integration with an advisory application within the intended institutional context.”

The Gap Bridge should naturally lead to the proposed study and demonstrate a clear progression:

- What is known → What remains insufficiently addressed → Why it matters
→ How the study responds → What the study contributes

This also ensures that the identified gap remains aligned with the study objectives:

- Data → Algorithms/Models → System/Application → Evaluation

Conceptual Framework

The Conceptual Framework presents the overall structure of the study by showing how its major components are logically connected. It provides a visual and textual representation of how the resources and requirements of the study are transformed through a series of research and development activities into the intended outputs.

For computing and technology-based studies, the conceptual framework should be directly aligned with the objectives of the study. Since the proposed studies follow the sequence:

Data → Algorithms/Models → System/Application → Evaluation
the conceptual framework should clearly reflect this progression.

For the examples in this guide, the Input–Process–Output (IPO) Model is used as the organizing framework. The IPO model presents the logical flow of the study through three major components: Input, Process, and Output.

The conceptual framework should not merely present a generic system development process. Each component should contain elements that are specific to the proposed study and its objectives.

Guide for Developing an IPO Conceptual Framework

Input

The Input component presents the essential resources, requirements, data, technologies, and other elements needed to conduct the study. These serve as the starting point of the research and development process.

Depending on the study, the inputs may include:



- Relevant datasets or data sources;
- User and system requirements;
- Hardware and software requirements;
- Selected algorithms or models for implementation and comparison;
- Technologies, frameworks, or development tools;
- Evaluation criteria and instruments.

Students should ensure that the inputs are consistent with the first and second objectives, particularly those involving data requirements and the selection of algorithms or technologies.

Process

The **Process** component presents the major activities undertaken to transform the inputs into the intended outputs. It should reflect the methodological and developmental procedures of the study.

For computing studies, the process may include:

- Data collection and preparation;
- Data preprocessing and analysis;
- Selection, implementation, training, or comparison of algorithms or models;
- Design and development of the system or application;
- Integration of technologies and system components;
- Testing and validation;
- Evaluation of algorithms or models;
- Evaluation of the developed system;
- Refinement and improvement.

The process should demonstrate the logical implementation of the study objectives.

A recommended sequence is:

Data Preparation → Algorithm/Model Implementation → System Development → Testing and Evaluation

Output

The **Output** component presents the expected results or deliverables of the study. These may include a developed system, application, model, framework, analytical results, or other research outputs.

For technology-based studies, the outputs should normally reflect both:

1. **The developed technological solution,** and
2. **The results of its evaluation.**

Examples include:

- A developed AI-powered application;
- A comparative analysis of implemented algorithms;
- Predictive or analytical results;
- An IoT-based monitoring system;

- Business Intelligence dashboards;
- System evaluation results;
- Recommendations for improvement.

The outputs should be directly connected to the **third and fourth objectives** of the study.

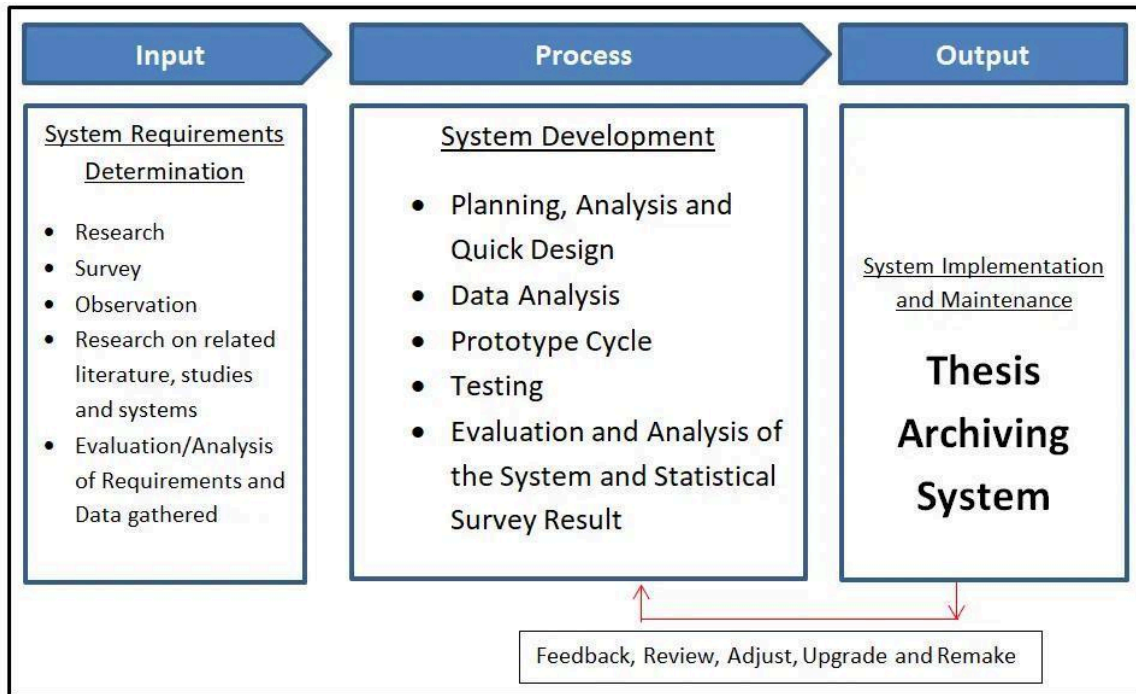
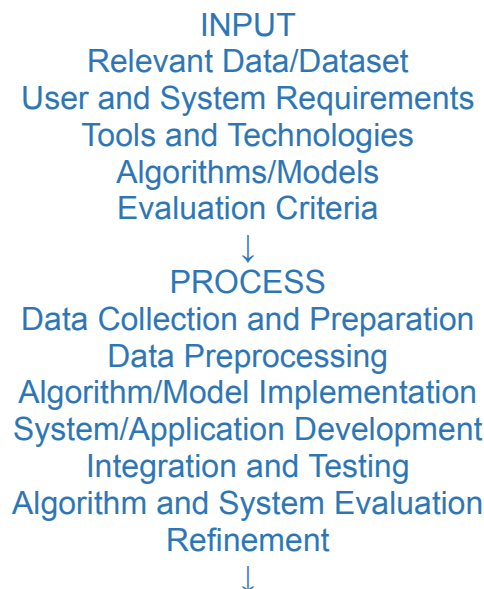


Figure X. Conceptual Framework



OUTPUT

Developed System/Application
Algorithm/Model Results
Analytical or Predictive Results
System Evaluation Results

EXAMPLE 1

Artificial Intelligence, Machine Learning, and Generative AI: AI-Powered Student Academic Risk Prediction and Advisory System

The conceptual framework of this study follows the Input–Process–Output (IPO) Model. It presents how relevant student academic data, selected Artificial Intelligence technologies, Machine Learning algorithms, Generative AI capabilities, system requirements, and evaluation criteria serve as inputs to the research process. These inputs undergo data preparation, algorithm implementation and evaluation, system development, technology integration, and testing to produce an AI-powered student academic risk prediction and advisory system.

IPO Components

Input

- Student academic data and relevant academic indicators;
- Data requirements and preprocessing requirements;
- User and system requirements;
- Selected Machine Learning algorithms, such as Logistic Regression, Random Forest, Support Vector Machine, and other appropriate models;
- Generative AI technology for generating advisory recommendations;
- Hardware and software resources;
- Algorithm evaluation metrics, including accuracy, precision, recall, and F1-score;
- System evaluation criteria, such as relevant ISO/IEC 25010 quality characteristics and usability measures.

Process

- Identification, collection, and preparation of relevant student academic data;
- Data cleaning, preprocessing, transformation, and feature selection;
- Implementation and training of selected Machine Learning algorithms;



- Comparative evaluation of the predictive models;
- Selection of the appropriate model based on the evaluation results;
- Design and development of the academic risk prediction and advisory system;
- Integration of the selected Machine Learning model and Generative AI advisory component;
- System testing and validation;
- Evaluation of the predictive models and the developed system;
- Refinement based on testing and evaluation results.

Output

- AI-Powered Student Academic Risk Prediction and Advisory System;
- Academic risk prediction results;
- Personalized AI-supported intervention recommendations;
- Comparative performance results of the implemented Machine Learning algorithms;
- System quality and usability evaluation results.

Example Diagram Content

INPUT	PROCESS	OUTPUT
Student academic data	Data collection and preparation	AI-Powered Student Academic Risk Prediction and Advisory System
User and system requirements	Data preprocessing and feature selection	Academic risk predictions
Selected ML algorithms	Model training and implementation	Personalized intervention recommendations
Generative AI technology	Algorithm comparison and evaluation	Algorithm performance results
Evaluation metrics	System design and development	System quality and usability results
Hardware and software resources	AI and Generative AI integration	
	Testing, evaluation, and refinement	

Narrative Explanation

The input stage consists of the essential elements required for the development of the proposed AI-powered student academic risk prediction and advisory system. The primary input includes relevant student academic data and



indicators that may be associated with academic performance and potential academic risk. These data will be examined and prepared according to the requirements of the selected predictive models. The inputs also include user and system requirements, appropriate hardware and software resources, and selected Machine Learning algorithms that may be implemented and evaluated for academic risk prediction. In addition, Generative AI technology will be considered for generating personalized and understandable advisory recommendations based on the system's predictive results. Appropriate algorithm performance metrics, including accuracy, precision, recall, and F1-score, together with relevant system quality and usability evaluation criteria, will also serve as inputs to the study.

The process stage involves transforming the identified inputs into the proposed system through activities aligned with the objectives of the study. Relevant academic data will first be identified, collected, cleaned, and preprocessed to prepare them for predictive modeling. Selected Machine Learning algorithms will then be implemented, trained, and evaluated to determine their performance in identifying students who may be academically at risk. The results of the model evaluation will guide the selection or implementation of the appropriate predictive approach. The academic risk prediction and advisory system will subsequently be designed and developed, integrating the predictive component with a Generative AI-supported advisory feature. The system and its components will undergo testing, validation, and



refinement. Both the predictive algorithms and the developed application will be evaluated using the identified performance and system quality measures.

The output stage consists of the developed AI-powered student academic risk prediction and advisory system and the results generated through its implementation and evaluation. The system is expected to provide academic risk prediction results and AI-supported intervention recommendations that can assist authorized users in understanding potential academic concerns. The output will also include the comparative performance results of the selected Machine Learning algorithms and the evaluation results of the developed system. These outputs will provide evidence regarding both the effectiveness of the implemented predictive models and the quality and usability of the proposed application.

EXAMPLE 2

Data Analytics and Business Intelligence: Data Analytics-Based Enrollment Forecasting and Decision Support System

The conceptual framework of this study follows the Input–Process–Output (IPO) Model. The framework illustrates how historical enrollment data, forecasting models, Business Intelligence technologies, user requirements, and evaluation criteria are processed through data preparation, forecasting model implementation, system development, and evaluation to produce an enrollment forecasting and decision-support system.

IPO Components

Input

- Historical enrollment data;
- Relevant institutional and program-related data;



- User and system requirements;
- Selected forecasting algorithms or models, such as ARIMA, SARIMA, Random Forest, or other appropriate approaches;
- Data Analytics and Business Intelligence tools;
- Hardware and software requirements;
- Forecasting evaluation metrics, such as MAE, RMSE, and MAPE;
- System quality and usability evaluation criteria.

Process

- Collection and preparation of historical enrollment data;
- Data cleaning, transformation, and exploratory analysis;
- Identification of enrollment patterns and trends;
- Implementation and training of selected forecasting models;
- Comparative evaluation of forecasting performance;
- Design and development of the decision-support system;
- Development of dashboards, visualizations, and reporting features;
- Integration of forecasting models and Business Intelligence components;
- System testing and validation;
- Evaluation and refinement.

Output

- Data Analytics-Based Enrollment Forecasting and Decision Support System;
- Enrollment forecasts;
- Historical trend analysis and visualizations;
- Interactive dashboards and reports;
- Comparative forecasting model results;
- System quality and usability evaluation results.

Example Diagram Content

INPUT	PROCESS	OUTPUT
Historical enrollment data	Data collection and preparation	Enrollment Forecasting and Decision Support System
Institutional data	Data cleaning and transformation	Enrollment forecasts
User and system requirements	Exploratory data analysis	Trend analysis
Forecasting models	Forecasting model implementation	Interactive dashboards
BI tools	Model comparison and evaluation	Reports and visualizations
Evaluation metrics	System development and integration	Forecasting performance results
Hardware and software resources	Testing, evaluation, and refinement	System quality and usability results



Narrative Explanation

The input stage includes the resources and information required for the development of the proposed Data Analytics-Based Enrollment Forecasting and Decision Support System. Historical enrollment data will serve as the primary source of information for identifying trends and developing forecasting models. Relevant institutional and program-related information may also be considered based on the availability and requirements of the study. Other inputs include the requirements of intended users, selected forecasting models, Data Analytics and Business Intelligence technologies, and the hardware and software resources required for implementation. Forecasting evaluation measures, including Mean Absolute Error, Root Mean Squared Error, and Mean Absolute Percentage Error, will be used to assess model performance. Relevant software quality and usability criteria will also guide the evaluation of the developed system.

The process stage begins with the identification, collection, and preparation of relevant historical enrollment data. The data will be cleaned and transformed before exploratory analysis is conducted to identify trends and patterns. Selected forecasting models will then be implemented and evaluated using appropriate forecasting error measures. The results of the evaluation will provide a basis for comparing the performance of the selected approaches. The system will subsequently be designed and developed to integrate forecasting capabilities with Business Intelligence functionalities. Dashboards, visualizations, and reporting features will be developed to enable authorized users to examine



historical information and projected enrollment results. The integrated system will undergo testing, validation, evaluation, and refinement.

The output stage consists of the developed Data Analytics-Based Enrollment Forecasting and Decision Support System and its corresponding analytical and evaluation results. The system is expected to generate enrollment forecasts and provide users with visualizations, dashboards, and reports that support the interpretation of historical and projected enrollment information. The outputs will also include the comparative performance results of the forecasting models and the results of the system quality and usability evaluation. Together, these outputs will demonstrate the performance of the analytical component and the practical applicability of the developed decision-support system.

EXAMPLE 3

Internet of Things, Embedded Systems, and Smart Technologies: Smart Campus Energy Monitoring and Predictive Control System Using IoT Sensors

The conceptual framework of this study follows the Input–Process–Output (IPO) Model. The framework illustrates how sensor-generated data, IoT devices, embedded technologies, predictive models, system requirements, and evaluation criteria are transformed through data acquisition, preprocessing, model implementation, system development, integration, testing, and evaluation into a smart energy monitoring and predictive control system.

IPO Components

Input

- Sensor-generated energy consumption data;
- IoT sensors and embedded devices;
- Relevant environmental or operational data;
- User and system requirements;



- Selected predictive algorithms or forecasting models;
- Communication and networking technologies;
- Hardware and software resources;
- Forecasting evaluation metrics, such as MAE, RMSE, and MAPE;
- System quality and usability evaluation criteria.

Process

- Installation and configuration of IoT sensors and devices;
- Collection and transmission of sensor-generated data;
- Data validation, cleaning, and preprocessing;
- Analysis of energy consumption patterns;
- Implementation and evaluation of selected predictive models;
- Design and development of the smart monitoring application;
- Integration of IoT devices, predictive models, and monitoring components;
- Development of dashboards and alerts;
- System testing and validation;
- Algorithm and system evaluation;
- Refinement and improvement.

Output

- Smart Campus Energy Monitoring and Predictive Control System;
- Real-time or near-real-time energy monitoring information;
- Energy consumption predictions;
- Monitoring dashboards and reports;
- Alerts or decision-support information;
- Predictive model performance results;
- System quality and usability evaluation results.

Example Diagram Content

INPUT	PROCESS	OUTPUT
Sensor-generated energy data	IoT data acquisition	Smart Energy Monitoring and Predictive Control System
IoT sensors and embedded devices	Data validation and preprocessing	Energy monitoring information
Environmental/operational data	Energy consumption analysis	Energy consumption predictions
Predictive models	Predictive model implementation	Dashboards and reports
Network technologies	Model evaluation	Alerts and decision support
User and system requirements	System development and integration	Model performance results
Evaluation criteria	Testing, evaluation, and refinement	System quality and usability results



Narrative Explanation

The input stage consists of the technological resources, data, and requirements needed to develop the proposed smart campus energy monitoring and predictive control system. The primary inputs include sensor-generated energy consumption data collected through IoT sensors and embedded devices. Relevant environmental or operational information may also be included when appropriate to the requirements of the predictive models. Other inputs consist of the user and system requirements, selected predictive algorithms, communication and networking technologies, and the hardware and software resources needed for system implementation. Appropriate forecasting performance measures, such as MAE, RMSE, and MAPE, will be used to evaluate the predictive models, while relevant software quality and usability criteria will be used to assess the developed application.

The process stage involves the transformation of sensor-generated data and technological resources into a functional smart monitoring system. IoT devices will be configured to collect and transmit relevant energy consumption information. The collected data will undergo validation, cleaning, and preprocessing to prepare them for analysis and predictive modeling. Selected algorithms or forecasting models will then be implemented and evaluated to determine their ability to estimate future energy consumption. The proposed system will subsequently be designed and developed by integrating IoT monitoring components, predictive models, data processing functions, and user-facing dashboards or reporting features. Testing and validation will be



conducted to examine the performance of both the predictive component and the overall application. The results will be used to refine and improve the system.

The output stage consists of the developed Smart Campus Energy Monitoring and Predictive Control System and the information produced through its operation and evaluation. The system is expected to provide energy monitoring information, energy consumption predictions, dashboards, reports, and relevant decision-support outputs. The output will also include the performance results of the selected predictive models and the results of the system quality and usability evaluation. These outputs will provide evidence regarding the effectiveness of the predictive approaches and the functionality and usability of the developed smart technology application.

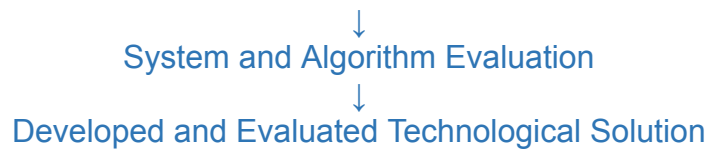
Important Alignment Guide

Students should ensure that the Conceptual Framework directly reflects the four specific objectives.

Objective	Conceptual Framework
Objective 1: Data	Input and early Process
Objective 2: Algorithms/Models	Input and Process
Objective 3: System/Application Development	Process and Output
Objective 4: Evaluation	Process and Output

The framework should therefore maintain the following logical progression:





Closing Paragraph for Chapter 2

Student Guide

The closing paragraph should briefly summarize the major insights obtained from the review, synthesis, and gap bridge. It should explain how the reviewed literature and studies support the proposed research without repeating the detailed discussions from the previous sections. The paragraph should establish that the identified research gap provides the basis for the proposed study and that the conceptual framework translates the findings into a logical research and development structure.

The closing paragraph should also create a smooth transition to the next chapter. It should not introduce new literature, technologies, or research claims that were not discussed in the review.

EXAMPLE 1: Artificial Intelligence, Machine Learning, and Generative AI

The review of related literature and studies establishes the growing importance of Artificial Intelligence, Machine Learning, and Generative AI in supporting data-driven decision-making in education. The reviewed works demonstrate that academic data can be used to identify patterns associated with student performance, while different Machine Learning algorithms provide varying levels of predictive performance depending on the characteristics of the dataset and implementation context. The literature also highlights the emerging potential of Generative AI to provide more understandable and personalized support based on analytical results.

However, the synthesis and gap bridge indicate the need for context-specific data preparation, comparative model evaluation, meaningful



integration of predictive and advisory technologies, and comprehensive evaluation of both the algorithms and the developed application. Guided by these findings, the proposed study will develop an AI-Powered Student Academic Risk Prediction and Advisory System structured through the Input–Process–Output framework. The succeeding chapter presents the research methodology and procedures that will be used to implement and evaluate the proposed study.

EXAMPLE 2: Data Analytics and Business Intelligence

The review of related literature and studies demonstrates the value of Data Analytics and Business Intelligence in transforming institutional data into information that supports planning and decision-making. Previous research shows that historical data and forecasting models can be used to identify trends and estimate future conditions, although model performance remains dependent on the characteristics of the available data and the implementation context. The reviewed works also emphasize the importance of integrating analytical results with dashboards, visualizations, and decision-support features to improve their practical usefulness.

The synthesis and gap bridge indicate the need for context-specific forecasting, comparative model evaluation, and the development of accessible systems that combine predictive analytics with Business Intelligence capabilities. These findings provide the foundation for the proposed Data Analytics-Based Enrollment Forecasting and Decision Support System. The next chapter



discusses the research methods, development procedures, and evaluation approaches that will be used in conducting the study.

EXAMPLE 3: Internet of Things, Embedded Systems, and Smart Technologies

The review of related literature and studies highlights the increasing role of Internet of Things technologies, embedded systems, and predictive analytics in the development of smart and data-driven environments. The reviewed works demonstrate that sensor-generated data can support continuous monitoring and that predictive models can extend traditional monitoring systems by providing estimates of future conditions. However, the synthesis and gap bridge emphasize continuing concerns regarding data quality, sensor reliability, connectivity, model selection, system integration, and comprehensive evaluation.

These findings demonstrate the need to investigate how IoT monitoring and predictive analytics can be effectively integrated within a context-specific smart technology application. Guided by the reviewed literature and the proposed Input–Process–Output framework, the study will develop a Smart Campus Energy Monitoring and Predictive Control System Using IoT Sensors. The succeeding chapter presents the methodology, development procedures, and evaluation methods that will guide the implementation of the study.

Insert Bibliography